BOUNDARY SIGNATURES, INCIDENCE RIGIDITY, AND GLOBAL CYCLE TRADES IN HAJÓS' CYCLE-DECOMPOSITION PROBLEM

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Abstract. For a finite simple graph with all degrees even, Hajós' cycle-decomposition conjecture asks whether its edges can be partitioned into at most $\lfloor(|V(G)|-1)/2\rfloor$ simple cycles. We develop two complementary, explicitly conditional tools for studying this open problem. First, for a cut consisting of an even matching, we give an exact composition formula for the minimum number of cycles, using boundary pairings and the vertex-intersection graph of the path components. For a four-edge matching cut this reduces to six integer costs on each side, and yields a sufficient inductive criterion with the exact parity-dependent vertex budget. Second, the cycle–vertex incidence graph controls the smallest support of a nontrivial cycle trade: when the incidence girth is $2h$ and distinct cycles meet in at most one vertex, the minimum support is exactly h. Applying a global three-edge-colouring construction and a transversal switch, we prove that the triangle projection of a cubic bipartite graph of girth at least 12 satisfies the sharp Hajós bound. We give a finite K_7 trade as a checkable example and audit a companion Lean 4 development against its literal theorem statements. The uploaded Lean code proves selected arithmetic and walk-surgery statements; it does *not* certify the unrestricted Hajós conjecture or the graph-theoretic theorems of this paper. We distinguish self-contained proofs, standard ingredients, computational checks, and unformalized claims throughout.

1. Introduction

Let G be a finite undirected simple graph with every vertex of even degree. A *cycle decomposition* is a family of pairwise edge-disjoint simple cycles whose union is $E(G)$; isolated vertices are permitted. Write

$$\operatorname{cyc}(G) = \min\{|\mathcal{C}| : \mathcal{C} \text{ is a cycle decomposition of } G\},$$
$$\operatorname{cyc}(G) = 0 \qquad \text{if } E(G) = \varnothing.$$

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For $n \geq 1$, let $B(n) = \lfloor (n-1)/2 \rfloor$. The (sharpened, equivalent) form of Hajós' conjecture is

$$\mathrm{cyc}(G) \leq B(|V(G)|). \tag{1}$$

The problem remains open in general. Its validity for graphs of order at most 12 was computationally established by Heinrich, Natale and Streicher [1]. Known structural results include planar graphs [4], graphs of maximum degree at most four (reviewed in [2, 3]), and graphs of pathwidth at most six [2]. In a different direction, the $O(n)$ cycle-and-edge decomposition conjecture of Erdős and Gallai was resolved in 2026 [16, 17]; that linear bound does not establish the constant $1/2$ in (1).

Our aim is not to replace these results or to claim a resolution of (1). We instead examine the precise information needed when local solutions are assembled into a global one, and when an inefficient cycle decomposition can be reorganized. The following points motivate the paper.

(i) Across a small cut, knowing only the minimum number of cycles on either side is insufficient. The pairing of boundary terminals and the possibility that distinct local paths meet at a vertex determine whether the global output is a genuine simple cycle rather than a closed trail.
(ii) The number of source cycles needed for the first nontrivial *trade* can grow arbitrarily large, even when the original graph is six-regular. A trade need not be a two-cycle recolouring.
(iii) A simultaneous global reorganization can succeed where all trades of a prescribed fixed support fail. We give a concrete sufficient condition for one natural class of six-regular triangle projections.

Our exact matching-cut formula and its four-port consequence are stated in Sections 3–4. Incidence rigidity and global reorganization appear in Sections 5–6. Section 8 gives a scope audit of the supplied Lean file; Section 9 discusses prior art and the remaining gap.

Provenance and priority. Matching decompositions, transfer states on small separators, colouring switches, and the use of incidence structures are established techniques. The results below are provided with independent mathematical arguments; we do not assert that every particular formulation is first in the literature. The attribution to the named earlier results is deliberate. A definitive priority or non-overlap statement would require a separate comprehensive bibliographic review and independent peer assessment.

2. Preliminaries and an elementary leaf-order bound

All paths in the paper are nonempty *simple paths* with two distinct endpoints; a simple cycle has at least three edges. Paths and cycles in an edge partition may share vertices, but not edges. We write $V(C)$ for the vertex set of a cycle C.

Lemma 2.1 (Leaf-ordered families)**.** *Suppose $C_1, \dots, C_k$, $k \geq 1$, are simple cycles that can be ordered so that for every $i < k$,*

$$\left| V(C_i) \cap \bigcup_{j>i} V(C_j) \right| \leq 1.$$

Then $\left|\bigcup_{i=1}^{k} V(C_i)\right| \geq 2k+1$. In particular, if these cycles form a cycle decomposition of an n-vertex graph, then $k \leq B(n)$.

Proof. Read the family in the reverse order. The first cycle uses at least three vertices. Every subsequently added cycle has at least three vertices and shares at most one with those already present, so contributes at least two new vertices. Hence the union has at least $3 + 2(k-1) = 2k+1$ vertices. The assertion for a graph follows because this union is a subset of $V(G)$. □

Remark 2.2. This elementary count applies, for example, to the natural cycle-block ordering of an Eulerian cactus. Such a count is not itself a new class-level result, nor does it prove that arbitrary Eulerian graphs admit this ordering. A version of the numerical argument appears among the supplied Lean lemmas; see Section 8.

3. Exact composition across a matching cut

Let $V(G) = A \dot\cup B$ be a vertex bipartition and suppose the cut $F = E_G(A, B)$ consists of $2r$ pairwise vertex-disjoint edges, with $r \geq 1$. Index these edges as

$$e_i = a_i b_i \quad (1 \leq i \leq 2r), \qquad a_i \in A,\ b_i \in B.$$

All a_i are distinct, and all b_i are distinct. Write $H_A = G[A]$ and $H_B = G[B]$. Since G has even degrees, the only odd-degree vertices in H_A are $a_1, \dots, a_{2r}$, and similarly on side B. The disjointness assumption on F is essential in what follows.

Definition 3.1 (Boundary state)**.** A *feasible local decomposition* of H_A is an exact edge partition into r nonempty simple paths and q simple cycles such that each a_i is an endpoint of precisely one path, and no other vertex is a path endpoint. The paths are allowed to intersect at other vertices, including at a terminal that is an interior vertex of another path.

It determines (a) a perfect matching π_A on $[2r]$, joining the indices of the endpoints of each path, and (b) the simple *path-intersection graph* X_A on these r paths, joining two path-vertices exactly when they share at least one vertex. We set

$$w_A(\pi_A, X_A) = \min\{q : \text{a feasible local decomposition has state } (\pi_A, X_A)\},$$

with $w_A = +\infty$ if the state is impossible. Define w_B analogously. The paths of a given matching are indexed by their endpoint pairs, so the intersection graph is unambiguous.

For two perfect matchings π_A, π_B on $[2r]$, their multiset union is a 2-regular multigraph $Q(\pi_A, \pi_B)$ on $[2r]$. A pair common to the two matchings gives a double edge in this *auxiliary* multigraph. Let $\kappa(\pi_A, \pi_B)$ be its number of connected components. Each component alternates between π_A-edges and π_B-edges. The double edges are permitted here because they represent two different paths joined through cut edges; they do not correspond to a 2-cycle of the original simple graph.

Call a pair of local states *compatible* when, in every component $S \subseteq [2r]$ of $Q(\pi_A, \pi_B)$, the distinct A-side paths indexed by the π_A-edges of S are pairwise vertex-disjoint, and the analogous condition holds on the B side. Equivalently, no two paths from the same component are adjacent in the corresponding X_A or X_B.

Theorem 3.2 (Exact matching-cut composition)**.** *Under the preceding hypotheses,*

$$\boxed{\operatorname{cyc}(G) = \min_{\substack{(\pi_A, X_A), (\pi_B, X_B) \\ \textit{compatible}}} \big[w_A(\pi_A, X_A) + w_B(\pi_B, X_B) + \kappa(\pi_A, \pi_B) \big].} \tag{2}$$

Every minimum is taken only over feasible states; in particular no conclusion is drawn from an incompatible pair.

Proof. Fix compatible decompositions on the two sides attaining their stated finite costs. Consider one connected component S of $Q(\pi_A, \pi_B)$. Follow its alternating π_A and π_B edges. Replace each π_A-edge ij by the corresponding simple path in H_A, and each π_B-edge by its path in H_B; insert the two appropriate cut edges at successive terminal labels. Compatibility means that, within S, the paths on each side have disjoint vertex sets. Since the vertex sets of A and B are disjoint, the resulting closed walk visits no vertex more than once except at its starting/ending point. Every path has at least one edge, so it is a genuine simple cycle, even when S is a two-vertex double-edge component. Different components use disjoint edge sets and may meet at vertices. The global cycles thus formed, together with the unchanged local cycles, partition $E(G)$ exactly; their number is $w_A + w_B + \kappa$. Taking a minimum proves the "$\leq$" direction.

Conversely, take any cycle decomposition $\mathcal{C}$ of G. Every cycle crosses F an even number of times. Delete the cut edges from every cycle that crosses F. The resulting components are nonempty simple paths in H_A or H_B: nonemptiness follows from the fact that distinct cut edges have distinct endpoints, so a cycle cannot make two consecutive cut-edge steps without an internal step. Every terminal is an endpoint of exactly one of these paths, as its unique cut edge is used exactly once. Thus each side contains exactly r paths. Cycles not meeting F are wholly inside one side and remain local cycles. A cross-cut cycle is simple, so the paths it uses on either side must have pairwise disjoint vertex sets. This gives a compatible pair of states. The cross-cut cycles correspond bijectively to components of $Q(\pi_A, \pi_B)$, and hence their number is κ. Consequently the number of members of $\mathcal{C}$ is at

least the corresponding right-hand side of (2). This proves "$\geq$" and the claim. □

Corollary 3.3 (Finite interface state space)**.** *The number of possible types* (π_H, X_H) *on one side is at most*

$$(2r-1)!!\, 2^{\binom{r}{2}}.$$

For $r = 2$ *this gives at most six state types, whereas for* $r = 3$ *it gives at most* $15 \cdot 8 = 120$*.*

Proof. There are $(2r-1)!!$ perfect matchings of $2r$ labels and at most $2^{\binom{r}{2}}$ simple graphs on the r path indices. Some formal states may be infeasible, which only reduces the count. □

Remark 3.4. The bound is on *interface state types*, not on the time required to compute their minimum costs. The exact formula does not imply a polynomial algorithm for unrestricted graphs or prove (1).

4. The four-edge cut and a sharp inductive budget

Now specialize Theorem 3.2 to $r = 2$. Denote the three perfect matchings of [4] by

$$p_1 = 12 \mid 34, \qquad p_2 = 13 \mid 24, \qquad p_3 = 14 \mid 23.$$

For $H \in \{H_A, H_B\}$ and a matching p, define

$$a_H(p) = \min\{q : \text{the two local paths realize } p \text{ and may meet}\},$$
$$b_H(p) = \min\{q : \text{the two local paths realize } p \text{ and are vertex-disjoint}\}.$$

Both costs include an arbitrary number q of local cycles, and equal $+\infty$ if the required decomposition does not exist. In particular, $a_H(p) \leq b_H(p)$. We emphasize that "may meet" is not a requirement that the paths intersect.

Theorem 4.1 (Exact four-edge interface)**.** *For an Eulerian simple graph with a four-edge matching cut,*

$$\boxed{\operatorname{cyc}(G) = \min\left\{ \begin{array}{l} \min\limits_{p\in\{p_1,p_2,p_3\}} \big(a_{H_A}(p) + a_{H_B}(p) + 2\big), \\ \min\limits_{p\neq q} \big(b_{H_A}(p) + b_{H_B}(q) + 1\big) \end{array} \right\}.} \tag{3}$$

Proof. If the two matchings agree, their union consists of two double-edge components. Each component yields a separate cross-cut cycle, so the two paths on the same side may meet: the crossing contribution is 2 and the optimal local cost is $a_{H_A}(p) + a_{H_B}(p)$. If the matchings differ, their union is one alternating four-cycle. Both paths from either side enter the *same* global cycle; hence they must be vertex-disjoint on each side, which is exactly what b_H records. The crossing contribution is 1. Applying Theorem 3.2 gives (3). □

The relevance to induction is made explicit by adding a temporary apex. For H_A, let H_A^+ be obtained by adding a fresh vertex z_A joined to a_1, a_2, a_3, a_4; define H_B^+ analogously. These graphs are simple and Eulerian and have $|A|+1$ and $|B|+1$ vertices, respectively. Put

$$m_A = \min_p a_{H_A}(p), \qquad m_B = \min_p a_{H_B}(p).$$

Lemma 4.2 (Rooting identity)**.** *With the above notation,*

$$\mathrm{cyc}(H_A^+) = m_A + 2, \qquad \mathrm{cyc}(H_B^+) = m_B + 2.$$

Proof. In any cycle decomposition of H_A^+, exactly two cycles contain the new vertex z_A, since it has degree four and a simple cycle uses exactly two edges incident to a vertex it visits. Deleting z_A from those two cycles leaves two nonempty simple paths whose endpoints partition the four terminals; every other cycle is unchanged. The number of residual cycles is the total number minus two, and is therefore at least m_A. Conversely, from any local decomposition realizing m_A with two paths, join the endpoints of each path to the new vertex. Because the new vertex is outside the old graph and all terminals are distinct, each path becomes a simple cycle; together with the m_A residual cycles they form a decomposition of H_A^+ into $m_A + 2$ cycles. The proof for B is the same. □

Theorem 4.3 (One-side flexibility suffices)**.** *Assume H_A^+ and H_B^+ both satisfy the Hajós bound. If for at least one side, say A, every matching $p \in \{p_1, p_2, p_3\}$ satisfies*

$$a_{H_A}(p) \le m_A + 1, \tag{4}$$

then G satisfies the Hajós bound.

Proof. Choose p attaining $a_{H_B}(p) = m_B$. By (3) and (4),

$$\mathrm{cyc}(G) \le a_{H_A}(p) + a_{H_B}(p) + 2 \le m_A + m_B + 3.$$

By Lemma 4.2 and the hypotheses,

$$m_A + 2 \le \left\lfloor \frac{|A|}{2} \right\rfloor, \qquad m_B + 2 \le \left\lfloor \frac{|B|}{2} \right\rfloor.$$

Consequently,

$$\begin{aligned} \mathrm{cyc}(G) &\le \left\lfloor \frac{|A|}{2} \right\rfloor + \left\lfloor \frac{|B|}{2} \right\rfloor - 1 \\ &\le \left\lfloor \frac{|A| + |B| - 1}{2} \right\rfloor = B(|V(G)|). \end{aligned}$$

The middle inequality holds for all four parity combinations of $|A|, |B|$, so the even-order boundary case is not discarded. □

Corollary 4.4 (A conditional restriction on a minimal counterexample)**.** *Let G be a counterexample of minimum order to (1). If G has a four-edge matching cut, then for each side $H \in \{H_A, H_B\}$ there is a pairing p for which either $a_H(p) = +\infty$ or $a_H(p) \ge m_H + 2$.*

Proof. The four distinct terminals on each side ensure $|A|, |B| \geq 4$, and thus $|A|+1 < |A|+|B|$ and $|B|+1 < |A|+|B|$. The smaller Eulerian graphs H_A^+, H_B^+ satisfy Hajós by the minimality assumption. If either side satisfied (4) for all pairings, Theorem 4.3 would contradict the choice of G. □

The exact residual budget can be written without a monotonicity guess. Assuming the rooted smaller graphs satisfy Hajós, set

$$\sigma_A = \left\lfloor \frac{|A|}{2} \right\rfloor - (m_A+2), \quad \sigma_B = \left\lfloor \frac{|B|}{2} \right\rfloor - (m_B+2), \quad P_\partial = \mathrm{cyc}(G) - m_A - m_B.$$

Both σ_A and σ_B are nonnegative integers.

Proposition 4.5 (Parity-exact boundary budget)**.** *In the above situation, G satisfies Hajós if and only if*

$$P_\partial \leq \begin{cases} 3+\sigma_A+\sigma_B, & |A| \text{ and } |B| \text{ are both even,} \\ 4+\sigma_A+\sigma_B, & \text{otherwise.} \end{cases} \tag{5}$$

Proof. Substitute $m_A = \lfloor |A|/2 \rfloor - 2 - \sigma_A$ and $m_B = \lfloor |B|/2 \rfloor - 2 - \sigma_B$ into $\mathrm{cyc}(G) = m_A + m_B + P_\partial$. The required inequality is then

$$P_\partial \leq 4 + \sigma_A + \sigma_B + \left\lfloor \frac{|A|+|B|-1}{2} \right\rfloor - \left\lfloor \frac{|A|}{2} \right\rfloor - \left\lfloor \frac{|B|}{2} \right\rfloor.$$

The final three terms equal -1 when both sizes are even, and 0 otherwise. This is exactly (5). □

Remark 4.6 (What the criterion does not say). There is no assertion that every Eulerian graph admits a four-edge matching cut, nor that every side has flexible pairings. In fact, independent computation of finite boundary signatures can produce costs differing by two or more; see Appendix A. It would be circular to apply Theorem 4.3 without verifying (4) or the rooted smaller-graph hypotheses.

5. Incidence forests and the minimum trade support

Let $\mathcal{C} = \{C_1, \ldots, C_k\}$ be a cycle decomposition of an Eulerian graph. Define its *cycle–vertex incidence graph* $\mathcal{I}(\mathcal{C})$ to have one bipartition class consisting of the cycle occurrences in $\mathcal{C}$, and the other consisting of vertices of G. A cycle-node C_i is adjacent to v if $v \in V(C_i)$. This records whether several cycles meet at the *same* vertex, a distinction lost in the ordinary intersection graph.

A *trade supported on* $S \subseteq \mathcal{C}$ replaces the cycles in S by another edge-disjoint family of simple cycles partitioning exactly $\bigcup_{C \in S} E(C)$, leaving every cycle outside S fixed. The support size is $|S|$. A trade is nontrivial when the multiset of edge sets of its replacement cycles differs from that of S; changing orientation, basepoint, or order does not count.

Lemma 5.1 (Incidence-forest rigidity)**.** *If $\mathcal{I}(\mathcal{C})$ is a forest, then $\mathcal{C}$ is the unique cycle decomposition of the graph with edge set $\bigcup_{C \in \mathcal{C}} E(C)$, up to ordering and orientation of cycles.*

Proof. Work in one nontrivial connected component of the incidence forest. Delete vertex-nodes of degree one while deciding which cycle-node to process first. The remaining bipartite forest, unless it consists of a single cycle-node, contains a cycle-node incident with at most one vertex-node that also lies on another cycle. Let C be such a cycle. The edges of C are connected to the remainder of the graph through at most one vertex x.

Every simple cycle containing an edge of C must remain inside C: at any other vertex of C there are exactly the two edges of C in the relevant edge union, while leaving C through x and subsequently returning to complete a closed traversal would visit x more than once. Consequently, the unique simple cycle containing any selected edge of C is C itself. Thus any cycle decomposition must include C. Delete its edges and induct on the number of cycle-nodes in the remaining incidence forest. Different connected components are independent, proving uniqueness. □

Theorem 5.2 (Exact first trade size)**.** *Suppose any two distinct cycles of $\mathcal{C}$ have at most one common vertex, and the finite girth of $\mathcal{I}(\mathcal{C})$ is $2h \geq 6$. Then*

(a) *no nontrivial trade has support smaller than h;*
(b) *a trade supported on exactly h cycles replaces them by two simple cycles, thereby decreasing the total number of cycles by $h-2$.*

In particular, the minimum support of a nontrivial trade, and of a cycle-count-decreasing trade, is exactly h.

Proof. If fewer than h cycle-nodes are selected, their incidence subgraph cannot contain any cycle: such a cycle would have length less than $2h$. By Lemma 5.1, their edge union has a unique cycle decomposition. This proves (a), even if an exchange is allowed to pass through intermediate decompositions with more cycles.

For (b), choose a shortest cycle in $\mathcal{I}(\mathcal{C})$. It alternates among h distinct cycle-nodes $C_1, \ldots, C_h$ and h distinct vertex-nodes $x_1, \ldots, x_h$, with C_i meeting C_{i+1} at x_i (indices modulo h). The selected cycles have no further intersections. Indeed, a second intersection between consecutive cycles would produce an incidence 4-cycle, impossible since $2h \geq 6$. An intersection between nonconsecutive cycles would add a length-two chord to the chosen shortest incidence cycle, producing a shorter incidence cycle. The same argument excludes any extra common vertex among selected cycles. Thus each C_i has two distinct designated intersection vertices and has no other vertex in common with any other selected C_j.

Split C_i at its two designated vertices into two internally disjoint nonempty arcs P_i, Q_i. The h arcs P_i join around the incidence ring to form a simple cycle; the h arcs Q_i join into a second simple cycle. Distinct source cycles intersect only at designated ends, so both outputs are simple. The two outputs are edge-disjoint and use every edge of the h inputs exactly once. Since $h \geq 3$, replacing h source cycles by these two cycles is a strictly decreasing trade, of support h. □

Corollary 5.3 (An obstruction to bounded-support repair)**.** *For every fixed integer $L \geq 1$, there is a finite simple six-regular Eulerian graph and a cycle decomposition $\mathcal{C}$ using more than $B(|V|)$ cycles, such that no nontrivial trade supported on at most L current cycles is possible. The same graph nevertheless satisfies Hajós.*

Proof. Choose a connected cubic bipartite graph $D = (P \dot\cup \mathcal{L}, F)$ of girth strictly larger than $\max\{2L, 12\}$. The existence of cubic graphs of arbitrarily large girth is classical; one may take a connected component of a bipartite double cover, which retains the necessary girth and cubic regularity (see [10]). Put $|P| = |\mathcal{L}| = n$. For each $\ell \in \mathcal{L}$, insert the triangle on its three neighbours in P. As D has no 4-cycle, these n triangles are edge-disjoint and form a simple six-regular graph $T(D)$; their incidence graph is D. Hence the given n-cycle decomposition is above $B(n)$, but its incidence girth is greater than $2L$. Lemma 5.1 applied to any set of at most L cycle-nodes proves the obstruction. The graph $T(D)$ satisfies Hajós by Theorem 6.4 below, since its incidence graph has girth at least 12. □

Remark 5.4. Corollary 5.3 refutes only algorithms that retain a valid cycle decomposition at every step and repartition the union of at most L *whole existing cycles*. It neither rules out graph reductions or path-based intermediate states, nor constitutes a counterexample to Hajós' conjecture.

6. Global reconstruction from cubic bipartite incidence graphs

Let $D = (P \dot\cup \mathcal{L}, F)$ be a finite *simple* cubic bipartite graph of girth at least six. For each $\ell \in \mathcal{L}$ create a triangle on its three neighbours in P. Write $T(D)$ for the resulting graph on vertex set P. Since D has no 4-cycle, distinct such triangles share at most one vertex and hence no edge. Every $p \in P$ lies in three triangles, so $T(D)$ is simple and 6-regular. The original $|\mathcal{L}| = |P| = n$ triangles form an edge decomposition, but are much too numerous for the Hajós bound.

A proper 3-edge-colouring of D exists by the perfect-matching theorem for regular bipartite graphs: extract a perfect matching, delete it, and colour the remaining 2-regular bipartite graph alternately with the other two colours. Label the three neighbours of ℓ by p_0, p_1, p_2 according to the colours of the incident edges ℓp_i. For $0 \leq i < j \leq 2$, assign the edge $p_i p_j$ in the triangle at ℓ to a spanning subgraph F_{ij} of $T(D)$.

Lemma 6.1 (Three spanning two-factors)**.** *The three graphs F_{01}, F_{02}, F_{12} form an edge partition of $T(D)$, and each is a spanning 2-regular graph. If D has girth $2h$, every cycle of each F_{ij} has length at least h. Consequently their aggregate number of cycles, denoted K, satisfies*

$$K \leq 3 \left\lfloor \frac{n}{h} \right\rfloor. \tag{6}$$

Proof. Each source triangle contributes its three edges to three different factors, so the factors partition $E(T(D))$. Fix $p \in P$. In the unique ℓ

incident with p by an edge of colour i, the triangle contributes one F_{ij}-edge at p; the unique colour-j incidence contributes a second. These two edges are distinct because a repeated neighbour would create a 4-cycle in D. Hence $d_{F_{ij}}(p) = 2$. In a finite simple graph, every component of a 2-regular graph is a simple cycle.

A cycle of the colour-i, j subgraph of D alternates between P and $\mathcal{L}$. Contracting each two-step P–$\mathcal{L}$–P passage to its assigned F_{ij}-edge yields a cycle of F_{ij}, and conversely. The former has exactly twice the length of the latter. Thus each F_{ij}-cycle has length at least h. Each factor covers all n vertices and consists of vertex-disjoint cycles, so it has at most $\lfloor n/h \rfloor$ components. Summing over the three factors proves (6). □

For girth at least 14, (6) immediately yields $K \leq 3n/7 < n/2$ and hence $K \leq B(n)$. Girth 12 requires a more exact argument, because the weak estimate can be exactly $n/2$.

Lemma 6.2 (Two shortest cycles in a cubic graph)**.** *Let D be a cubic simple graph of girth g. If C and C' are distinct cycles of length g and intersect, then their common edges form one nonempty path. In particular, in a properly 3-edge-coloured cubic graph, two such cycles using distinct pairs of colours have at most one common edge.*

Proof. At a common vertex, two cycles of a cubic graph must share an incident edge, since two 2-element subsets of a 3-element set have a common member. Thus the intersection cannot consist only of isolated common vertices. Suppose its common-edge subgraph has $t \geq 2$ path components. Along C', consecutive common path components are joined by t internally C-disjoint path segments, each of positive length. The t common paths have at least t edges in total, so the shortest of these complementary segments has length at most $(g - t)/t < g/2$. Its endpoints are distinct vertices on C. Join them by a shorter arc of C, of length at most $g/2$. The union of these two internally disjoint paths is then a simple cycle of length strictly less than g, a contradiction. Thus there is exactly one common path component.

For the last assertion, two cycles with different colour pairs have only one colour in common; every common edge has that colour. Proper edge-colouring forbids two consecutive edges of the same colour, so a nonempty common path contains at most one edge. □

Lemma 6.3 (Transversal bichromatic switch)**.** *Suppose D is properly edge-coloured with colours $0, 1, 2$ and C is a $0, 1$-alternating cycle of length $2r$. Assume its r colour-0 edges belong to r distinct $0, 2$-alternating cycles, and its r colour-1 edges belong to r distinct $1, 2$-alternating cycles. Exchanging colours 0 and 1 along C produces another proper 3-edge-colouring for which the aggregate number of projected factor cycles decreases by exactly $2r - 2$.*

Proof. The colour-$0, 1$ factor is unchanged. Consider the r old $0, 2$ cycles touched by C. By hypothesis, each contains exactly one of the specified colour-0 edges, and the cycles are pairwise vertex-disjoint. Deleting those

edges leaves r pairwise vertex-disjoint paths. Under the switch, the former colour-1 edges of C become colour-0 edges and join these r paths cyclically into a single simple $0,2$ cycle. Therefore this factor loses $r-1$ components. The same argument, with the colours 0 and 1 interchanged, shows that the $1,2$ factor loses $r-1$ components. All untouched components remain untouched, giving a total loss of $2r-2$. □

Theorem 6.4 (A girth-12 sufficient condition)**.** *Let $D = (P \dot\cup \mathcal{L}, F)$ be a finite simple cubic bipartite graph with girth at least 12, and let $T(D)$ be its triangle projection. Then*

$$\boxed{\operatorname{cyc}(T(D)) \le \left\lfloor \frac{|P|-1}{2} \right\rfloor .}$$

Proof. Put $n = |P|$ and choose any proper 3-edge-colouring of D. By Lemma 6.1, every projected factor cycle has length at least six. Since the three 2-factors have $3n$ edges in total, $6K \le 3n$, or $2K \le n$.

If $K \le B(n)$, we are done. Otherwise the integrality of K forces n even and $K = n/2$. Equality in $6K \le 3n$ means *every* projected factor cycle has length exactly six. Equivalently, all bichromatic cycles of D have length exactly 12, and the girth is exactly 12.

Choose any bichromatic $0,1$ cycle C of length 12. For each of its six colour-0 edges, let D_i be the $0,2$ bichromatic cycle containing that edge. Every D_i is a girth cycle of length 12. If two of those edges belonged to the same D_i, then C and D_i would have two common edges, contradicting Lemma 6.2. Thus the six D_i are distinct. The same reasoning applies to the six colour-1 edges and the $1,2$ bichromatic cycles. Therefore C satisfies the transversal hypotheses of Lemma 6.3 with $r = 6$. Switching its two colours yields a new valid three-factor decomposition with

$$K' = K - 10 \le n/2 - 1 = B(n).$$

These are actual simple cycles and a complete exact edge partition, so the required Hajós decomposition exists. □

Remark 6.5 (What is new in the argument, not in its ingredients). The use of Hall's theorem to three-edge-colour a cubic bipartite graph, the existence of large-girth cubic graphs, and Kempe switching are standard. The preceding proof isolates an exact-budget application to the triangle projection $T(D)$ and closes the potentially problematic equality case by a transversal switch. We do not claim bibliographic priority for this particular corollary, nor extend it to arbitrary six-regular graphs.

7. A finite trade and computational checks

The graph-theoretic results above do not depend on numerical search. For transparency, we record one short example directly related to the supplied Lean file.

Example 7.1 (A four-to-three trade in K_7)**.** Take vertices $0, 1, \ldots, 6$. The following four cycles partition the edges of K_7:

$$T = (0, 1, 2),$$
$$C_1 = (1, 3, 2, 4, 5, 6),$$
$$C_2 = (0, 3, 4, 6, 2, 5),$$
$$C_3 = (0, 4, 1, 5, 3, 6).$$

Repartition the same 21 edges into the three Hamilton cycles

$$H_1 = (0, 3, 1, 6, 2, 4, 5),$$
$$H_2 = (0, 1, 2, 3, 5, 6, 4),$$
$$H_3 = (0, 2, 5, 1, 4, 3, 6).$$

Each list has distinct vertices, each displayed consecutive pair is an edge of K_7 (with the closing edge understood), and a direct set check shows that the cycles are pairwise edge-disjoint in each decomposition and that both families cover the 21 edges. The source file proves the *equality of the two union edge sets* by the finite tactic `decide`; pairwise edge-disjointness and the simple-cycle semantics are separate mathematical checks, not the literal content of that Lean statement. The final count $3 = B(7)$ is sharp because the complete graph has degree six at each vertex.

Remark 7.2 (Computational evidence is not a new finite verification record)*.* The supplementary files also contain an explicit eight-cycle partition of one 19-vertex Eulerian graph with maximum degree 16, together with its input edge set. Edge coverage proves an upper bound of eight and degree 16 proves the matching lower bound; this certifies $\mathrm{cyc}(G) = 8$ *for that graph only.* Earlier exhaustive verification of all Eulerian graphs through order 12 is due to [1] and is not repeated here as a novel computational claim.

8. What the companion Lean file establishes

The supplied file `HajosLemmas.lean` is a Lean 4 file using the Mathlib `SimpleGraph.Walk` interface [14, 15]. Its edge type is `Sym2 V`, so it models undirected edges, and its cycle type is based on the genuine predicate `SimpleGraph.Walk.IsCycle`. The decomposition predicate combines no repeated occurrences in the concatenated edge list with exact membership equivalence to the graph's edge set.

The file declares the unrestricted conjecture as a proposition, in schematic Lean form,

```
def HajosConjecture :  Prop :=
    forall (W : Type u) [Fintype W] (G : SimpleGraph
W),
      IsEvenGraph G -> HajosBound G
```

It is a `def`, *not* a theorem accompanied by a proof term. The distinction is mathematically fundamental: declaring a proposition does not establish it.

TABLE 1. Scope of the submitted Lean source. "Script present" is not an independent kernel-verification claim.

Source declaration(s)	Literal content and limitation
`IsLeafOrdered.` `hajos_count`; `leaf_ordered_` `decomposition_hajos`	A numerical Hajós bound under an explicitly supplied leaf-ordering and, for the latter declaration, an explicitly supplied valid decomposition. Not a theorem that every Eulerian graph has the ordering.
`close_path_isCycle`; `subdivide_based_` `cycle`; `contract_path_isCycle`	Local genuine-walk constructions under stated path, adjacency and surviving-edge hypotheses. Not an unrestricted decomposition theorem.
`kempe_sufficient_` `decrease`; `three_factor_budget`	Arithmetic implications *assuming* the relevant combinatorial counts. The matching, projection and exchange constructions are not derived from these statements.
`K7_trade_` `preserves_edges`	Finite equality of two unions of undirected edge sets in the displayed K_7 example. It does not itself assert no duplicate edges or prove the global conjecture.
`HajosConjecture`	Definition of the universal target proposition, not a proved result.

A static inspection of the supplied 546-line source found no use of `sorry` as a proof placeholder and no user-declared `axiom`. The file does contain text explicitly calling portions of itself "source draft" and "not compiled or kernel-checked". The public repository currently includes the source file but does not include a pinned Lean project configuration or a continuous-integration log from which we can independently reproduce the assertion that the entire file passed. The author may possess a successful private build; this manuscript does not convert that assertion into a reproducible kernel-check claim.

Crucially, Theorems 3.2, 5.2, and 6.4 are *proved in mathematical prose in this paper* and are *not* asserted to be formalized in the supplied Lean file. A complete mechanization would need typed local path decompositions, transport between graphs, endpoint-pairing compatibility, the global incidence argument, and the three-colour switching theorem.

9. Discussion, limitations, and open problems

The incidence and cut approaches address different failures of overly local arguments. In the matching-cut setting, even an optimal local number of cycles loses information about how exposed paths connect. For incidence-based trades, a cycle decomposition may be completely rigid under all changes of bounded support, yet admit a much more economical global reorganization. These facts make a simple descent algorithm based on arbitrary two-cycle exchanges inadequate as a general proof strategy.

Relation to established work. The four-edge signature is a specialized boundary-state construction; it should be compared with classical dynamic programming on graph separators and with the path-reconnection and recolouring techniques of [2, 7]. The incidence approach is related to previous investigations of the structure and possible cardinalities of cycle decompositions [6]. We use Hall's perfect-matching theorem and standard three-edge-colouring/switching arguments rather than claiming them as innovations. The large-girth cubic-graph existence input belongs to the classical literature [10]. The present work does not reprove the planar, low-degree, low-pathwidth or dense-graph theorems of [4, 2, 11, 12, 13].

A finite-state problem with genuine content. In Theorem 4.3, two rooted smaller graphs satisfy the sharp bound, but this alone does not guarantee a sharp bound for the glued original graph. A natural next problem is to establish a bound linking the variation of $a_H(p)$ over terminal pairings to structural features that also create enough vertex-budget slack in Proposition 4.5. This problem is not settled here.

High connectivity. Theorems 3.2 and 4.3 require a matching cut. They give no direct reduction when the graph has no such small cut. Likewise, Theorem 6.4 deals only with triangle projections of cubic bipartite incidence graphs of sufficiently large girth, and not all six-regular graphs.

No solution of the unrestricted conjecture. The independent statements in this paper do not combine to prove (1) for arbitrary Eulerian graphs. Nor does the construction of a counterexample to a *trade algorithm* produce a counterexample to Hajós' conjecture. Establishing a global inequality for the interface costs or a valid operation for highly intersecting decompositions remains necessary.

Data and code availability

The user-supplied Lean source file is publicly readable at https://github.com/AAAAAlbert1958/Hajos-Problem (source-file Git blob `559bbde0d523bb80a0be39236707013adf761187`). Its copy is included with this manuscript. The accompanying source archive also contains selected Python scripts, input data, and independently checkable finite certificates from the preceding computational investigations. At the time this manuscript was prepared, the public GitHub listing did *not* yet contain the Python scripts; therefore we do not claim they are already hosted there. They should be deposited at the same stable repository (ideally at an immutable release tag) before a submission states otherwise. None of the principal mathematical theorems requires trust in those scripts.

Acknowledgments

Appendix A. A finite boundary-signature stress test

This appendix documents one finite instance that demonstrates why a four-edge interface cannot be summarized by a single minimum local cost.

The example is *computational diagnostic evidence*, not a new class theorem for all 24-vertex graphs.

Let H_A have vertex set $\{0, \dots, 11\}$, terminal order $(1, 7, 9, 11)$, and the following 16 edges:

$$E(H_A) = \{(0,7),(0,11),(1,3),(1,5),(1,10),(2,5),(2,6),(3,4),\\ (3,6),(3,9),(4,10),(5,6),(5,7),(6,8),(7,11),(8,11)\}.$$

Let H_B be a copy with all labels increased by 12. Use terminal order $(13, 19, 23, 21)$, and add the four independent edges

$$(1,13), \quad (7,19), \quad (9,23), \quad (11,21).$$

The resulting graph has 24 vertices and 36 edges, all degrees even. The three boundary pairings in the global terminal order have locally computed costs

	p_1	p_2	p_3
a_{H_A}	1	2	0
b_{H_A}	3	2	∞
a_{H_B}	1	0	2
b_{H_B}	3	∞	2

Formula (3) therefore gives $\mathrm{cyc}(G) = 4$: all three same-pairing choices cost four, while the least feasible different-pairing choice costs five. The two sides each have $m_H = 0$, so the boundary penalty is $P_\partial = 4$, disproving the tempting *auxiliary* inequality $P_\partial \le 3$ without a slack or flexibility assumption.

For independent finite verification, one may enumerate at each even-degree vertex all pairings of incident edges; at each odd terminal one incident edge remains unpaired. The induced trails are accepted only if they form the specified simple paths and simple cycles. This exhaustive transition enumeration gives 729 local systems per side and, for the entire graph, $3^{12} = 531441$ transition systems, of which 1337 correspond to valid simple-cycle decompositions. Their cycle-count distribution is

number of cycles	4	5	6	8
valid transition systems	560	400	376	1.

This independent whole-graph test is consistent with the interface formula. Complete input edges and checker source are supplied in the accompanying archive. The example is far below the Hajós bound $B(24) = 11$ and is *not* a counterexample to the conjecture.

Appendix B. Formalization checklist

To make a future statement of machine verification independently reproducible, a source release should include:

1. an immutable Mathlib revision and matching Lean toolchain declaration;
2. a buildable Lean project file and a clean `lake env lean` invocation or an equivalent successful CI run;

3. the literal universal theorem statement (when and if proved), rather than only a definition of `HajosConjecture`;
4. full theorem dependencies and `#print axioms` output, with an explanation of any additional axioms;
5. typed edge-disjointness and edge-covering witnesses for examples, not merely equality of their edge-set unions;
6. formal proofs of all theorems expressly described as mechanized in the paper, with unformalized parts clearly marked.

A lack of `sorry` by itself is not a substitute for a successful kernel check, and a Lean theorem with extra mathematical hypotheses must not be advertised as an unconditional proof.

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